



Reflect with us - Week 3



This assignment will not be graded and is only for practice.

1 point

Choose the set of correct options.

- ☐ Any subset of any linearly independent set of vectors is linearly independent.
- ☐ Any subset of any linearly dependent set of vectors is linearly dependent.
- ☐ Any subset of any linearly dependent set of vectors is linearly independent.
- ☐

Any set of 33 vectors in R^3 is linearly independent.



Any set of 44 vectors in R^3 is linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Any subset of any linearly independent set of vectors is linearly independent.

Any set of 44 vectors in R^3 is linearly dependent.

Recall the definition of linearly dependent and independent:

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Statement 1) Definition of linear dependence: A set of vectors $\{v_1, v_2, \dots, v_n\}$ in V is said to be linearly dependent if there exist scalars $a_1, a_2, \dots, a_n \in \mathbb{R}$, not all zero such that $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$.

Statement 2) Definition of linear independence: A set of vectors $\{v_1, v_2, \dots, v_n\}$ in V is said to be linearly independent if $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$ implies that $a_i = 0$ for $i = 1, 2, \dots, n$.

Statement 3) If S is a set containing zero vector, then the set S is a linearly dependent set.

Statement 4) Let S be a subset of a vector space V . If an element of S is a scalar multiple of another element from the set S , then S is a linearly dependent set.

Statement 5) If S is linearly dependent, then there exists $v \in S$, such that v can be written as a linear combination of other vectors in S and vice versa.

Discussion on Option 1: Let $S = \{v_1, v_2, \dots, v_n\}$ be a linearly independent set of vectors and let S' be a subset of S .

Claim: S' is linearly independent.

1 point

What we have to assume for S' if we want to use the method of contradiction?



Let us assume that S' is linearly dependent.



Let us assume that S' is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let us assume that S' is linearly dependent.

If S' is linearly dependent, then there exists $v \in S'$, such that v can be written as a combination of other vectors in S' .



Which of the above statements we are using here?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

$v \in S' \vee v \in S'$ implies $v \in S \vee v \in S$, as $S' \subseteq SS' \subseteq S$. Those vectors of whose linear combination is giving $v \vee v$ are also in SS due to the same argument.

1 point

Again from statement 5, can we conclude that SS is linearly dependent?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

But SS was given to be a linearly independent set, hence we arrive at a contradiction. So our claim is proved.

Another method: Another method: Let $S' = \{v_{i_1}, v_{i_2}, \dots, v_{i_k}\} \subseteq SS' = \{v_{i_1}, v_{i_2}, \dots, v_{i_k}\} \subseteq S$.

Consider the equation

$$a_1 v_{i_1} + a_2 v_{i_2} + \dots + a_k v_{i_k} = 0 \quad a_1 v_{i_1} + a_2 v_{i_2} + \dots + a_k v_{i_k}$$

= 0

1 point

Can we conclude each $a_i = 0$ $a_i = 0$ from here?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

As all the v_{i_j} v_{ij} are also in SS , the given equation is a linear combination of vectors from SS . Now as SS is linear independent, $a_i = 0$ $a_i = 0$ for each $i \in \{1, 2, \dots, k\}$ $i \in \{1, 2, \dots, k\}$. Hence $S' S'$ is linearly independent.

∴ Now try Question 7 and 8 of activity 3.3 Now try Question 7 and 8 of activity 3.3

Discussion on Option 2: Discussion on Option 2: Let us discuss an example. Let

$S = \{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ $S = \{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ be a subset of R^3 R^3 .

1 point

Is SS linearly dependent?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: $(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$

Accepted Answers:

Yes.

1 point

Can you find a linearly independent subset of SS ?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: Check that any proper subset of SS of cardinality 1 or 2 is linearly independent subset of SS .

Accepted Answers:

Yes.

Consider $S' = \{(1, 0, 0), (0, 1, 0)\}$ $S' = \{(1, 0, 0), (0, 1, 0)\}$.

1 point

Is $S' \subseteq SS' \subseteq S$?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

1 point

Is $S' S'$ linearly independent?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Hence Option 2 is not correct.

Discussion on Option 3: Discussion on Option 3: Here also we construct an example. Let

$S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$ $S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$.

1 point

Is SS linearly dependent?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Consider $S' = \{(1, 0, 0), (2, 0, 0)\}$ $S' = \{(1, 0, 0), (2, 0, 0)\}$.

1 point

Is $S' \subseteq SS' \subseteq S$?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

1 point

Is $S' S'$ linearly independent?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

Hence Option 3 is not correct.

Discussion on Option 4: Discussion on Option 4: Consider the same set $S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$
 $S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$ that we have discussed previously.

1 point

Is SS linearly independent?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

1 point

What can we conclude about Option 4 from this?

☐ Option 4 is correct.

☐ Option 4 is not correct.

☐ Nothing can be concluded about Option 4 from here.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Option 4 is not correct.

· · Now try Question 5 and 8 of activity 3.4 Now try Question 5 and 8 of activity 3.4

Discussion on Option 5: Discussion on Option 5: Let $S = \{v_1, v_2, v_3, v_4\}$ $S = \{v_1, v_2, v_3, v_4\}$ be a set of 4 vectors in R^3 . Let $v_1 = (x_1, y_1, z_1)$ $v_1 = (x_1, y_1, z_1)$, $v_2 = (x_2, y_2, z_2)$ $v_2 = (x_2, y_2, z_2)$, $v_3 = (x_3, y_3, z_3)$ $v_3 = (x_3, y_3, z_3)$, $v_4 = (x_4, y_4, z_4)$ $v_4 = (x_4, y_4, z_4)$. Consider the following equation,

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0 \quad a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0$$

i.e.,

$$a_1(x_1, y_1, z_1) + a_2(x_2, y_2, z_2) + a_3(x_3, y_3, z_3) + a_4(x_4, y_4, z_4) = (0, 0, 0)$$

$$a_1(x_1, y_1, z_1) + a_2(x_2, y_2, z_2) + a_3(x_3, y_3, z_3) + a_4(x_4, y_4, z_4) = (0, 0, 0)$$

i.e.,

$$\begin{aligned} a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4 &= 0 \\ a_1 y_1 + a_2 y_2 + a_3 y_3 + a_4 y_4 &= 0 \\ a_1 z_1 + a_2 z_2 + a_3 z_3 + a_4 z_4 &= 0 \end{aligned}$$

$$+ a_3 y_3 + a_4 y_4 + a_1 z_1 + a_2 z_2 + a_3 z_3 + a_4 z_4 = 0 = 0 = 0$$

The matrix representation of this system of linear equations is given by

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ z_1 & z_2 & z_3 & z_4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We want to find the solution for the vector $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$ for this given homogeneous system of

linear equations.

1 point

What can we say about the solutions of this system of linear equations?

- ☐ The solution is unique.
- ☐ There are infinitely many solutions.
- ☐ There must exist a non-zero solution.
- ☐

00 is the only solution.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: Try to compute the maximum number of dependent variables you can get from the row reduced echelon form of the above matrix.

Accepted Answers:

There are infinitely many solutions.

There must exist a non-zero solution.

1 point

Is SS linearly independent?

- ☐ Yes.
- ☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

Hence Option 5 is correct.

· · This will help you to solve Question number 4 of the graded assignment.
This will help you to solve Question number 4 of the graded assignment.

[Check Answers](#)

Your score is: 0/16